

Group name

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Section

“Morals excite passions, and produce or prevent actions. Reason of itself is utterly impotent in this particular. The rules of morality, therefore, are not conclusions of our reason.” (David Hume, *A Treatise of Human Nature*, 1739)

ہر خاکی و نوری پہ حکومت ہے خرد کی

باہر نہیں کچھ عقل خدا داد کی زد سے

عالم ہے غلام اس کے جلال ازلی کا

اک دل ہے کہ ہر لحظہ الجھتا ہے خرد سے

اقبال

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

Read the passage below, making note of any important points and definitions.

A well-constructed scientific argument, defending a scientific conclusion, generally rests upon two foundations: reliable empirical evidence and sound logical arguments. Of course there have been scientific arguments that weren't based on good evidence and reasoning, but these shoddy arguments (and the conclusions based on them) generally don't withstand the test of time. The point isn't that we always have proper evidence and reasoning in the sciences, the point is that we always should have these things. This point is not trivial. In some areas of human thought, conclusions may quite properly not be based on logic and evidence. In some political discourse, for example, we might legitimately based our conclusions on a shared set of values and traditions instead of rational analysis (elected government versus divine right of kings, for instance). Although such considerations sometimes enter into scientific thinking, science still provides an excellent example of reasoned discourse. As such, we can at least use science as a starting point for a discussion of valid argumentation across the board.

The subject matter of this chapter, as you see, extends far outside the boundaries of science. We are concerned here with methods of clear thinking and critical analysis that are relevant to any issues. Where the methods and thought processes typical of the sciences are applicable, these methods are invaluable. When issues turn on differences in values, faith, cultural background, and so on, then we are obligated to isolate these differences and identify them clearly. Differing values are no excuse for bad logic and lack of evidence. Valid argumentation in the murky and ambiguous issues of human affairs is not essentially different from valid argumentation in the esoteric realms of science; it's just more difficult.

THE CONTRIBUTIONS OF SCIENCE

What does critical analysis of argumentation have to do with science? One simple answer to this question is that science routinely employs this very kind of critical analysis all the time. Science is one of the few human endeavors

in which we sometimes have the luxury of starting with a general premise, deductively working out the results of this premise, and comparing these results to evidence of extremely high quality. Examining this process at work in the relatively tidy and uncomplicated problems of science gives us a sense of how to proceed in the more difficult realms of politics, economics, social issues, and so on. The skeptical attitude characteristic of the sciences also fosters a spirit of critical analysis across a wide range of issues. Beyond these general considerations, however, science offers several techniques and modes of thinking that are not commonly found in other fields.

USING BASIC KNOWLEDGE

Although science continues to progress as new results are discovered and old theories are modified, we have a certain amount of core knowledge and experience in science that is not likely to change radically or quickly. Unlike the trends and fads of political ideologies and public opinion, we can count on this basic scientific knowledge to be correct. When such knowledge is relevant to the premises of an argument, then we can evaluate these premises without knowing every factual detail. An example is the conservation of energy law. There are no known exceptions to this principle, and none are expected. If the success of some public policy initiative depends on the creation of energy from nothing, then we can reject this policy without further consideration. You don't need an extensive stock of scientific knowledge to apply this kind of reasoning. For example, it's clear on very fundamental grounds that the earth is finite. Anyone who argues that the earth's resources are limitless must then be wrong. Any argument that depends on limitless resources as a premise is likewise wrong. Another example of basic knowledge is the form and properties of exponential growth. Whether the subject of debate is population growth, economics, resource use, finance, or energy policy, any exponentially changing quantity shares the same properties and characteristics; and these properties are not subject to con-

trary opinion, they are simply matters of arithmetic. If the premise of an argument demands a contradiction of these known properties, then you may safely conclude that this premise is wrong. Notice that both of these examples provide us with constraints on truth, giving information

about what must be wrong instead of telling us that something is right.

Excerpted from Gregory N Derry. *What science is and how it works*. Princeton University Press, 1999

Reflect and discuss (A)

 1.

 2.

 3.

 4.

- | | |
|---|--|
| a) On a baseball field, the center of the pitcher's mound is 59 feet from home plate. | b) Green is the most pleasant color to look at. |
| c) Driving while drowsy is dangerous. | d) Pit vipers can strike a warm-blooded animal even when it is pitch dark. |
| e) Donald Trump is overweight | f) This lemon tastes great because it isn't bitter. |
| g) There is life on Mars. | h) Your teacher will complain if you text in class. |

ctd. on next page

- i) Your teacher would be crazy not to complain if you text in class.
- j) God exists
- k) Donald Trump is fat

Read the passage below, making note of any important points and definitions.

DETERMINING THE KIND OF CLAIM YOU SHOULD MAKE

The kind of problem you pose determines the kind of claim you make and the kind of argument you need to support it. Academic researchers usually pose not *practical* problems but *conceptual* ones, the kind whose solution asks readers not to act but to understand:

“The Great Recession of 2008–2011 was caused largely by negligent financial regulation that let banks take on too much risk.”

MAKING CONCEPTUAL CLAIMS

Beyond distinguishing between practical and conceptual claims, it is useful to recognize that claims address a range of questions: Does a thing or a situation exist? If so, how should we characterize it? How did it get this way? Is it good or bad? What can or should be done about it? Depending on the question or questions implicit in a claim, different kinds of evidence will be required to support it. The claims you make on a topic will fall into one (or more) of the following classes:

1. Claims of fact or existence: Average global temperatures have risen to unprecedented levels within the past decade.
2. Claims of definition and classification: Birds, not reptiles, are the direct descendants of dinosaurs.
3. Claims of cause and consequence: Exposure to asbestos is a leading contributor to lung cancer.
4. Claims of evaluation or appraisal: Shakespeare’s greatest comedy is *As You Like It*.

5. Claims of action or policy: Pennsylvania should increase extraction fees on natural gas drilling to fund education.

MAKING PRACTICAL CLAIMS

A practical claim is one that argues for (or against) some action or policy. It is usually built from a chain of conceptual claims: one that demonstrates that a problem exists, another that shows what causes the problem, and still another that explains how doing what you propose will fix it. Readers may also expect you to explain the following:

1. Why your solution is feasible; how it can be implemented with reasonable time and effort.
2. Why it will cost less to implement than the cost of the problem.
3. Why it will not create a bigger problem than the one it solves.
4. Why it is cheaper or faster than alternative solutions – a claim often difficult to support.

If you advance a practical claim but don’t make those four sub-arguments, your readers may reject your whole argument.

Excerpted from Wayne C et al. Booth. *The craft of research*. University of Chicago press, 2016

Reflect and discuss (B)

 5.

 6.

1. TV inflates estimates of crime rates.

2. Graphic reports of violence on local TV news lead regular viewers to overestimate by as much as 150 percent both the rate of crime in their neighborhood and the personal danger to themselves and their families.
3. The Forest Service has spent a great deal of money to prevent forest fires, but there is still a high probability of large, costly ones.
4. “A recession is a significant decline in economic activity spread across the economy, lasting more than a few months, normally visible in real gross domestic product, real income, employment, industrial production and wholesale-retail sales.” (National Bureau of Economic Research, USA)

 7.

 8.

 9.

1. If you’ve ever met a pet bird, you know they are busy creatures.
2. As a long-time customer, you’re already taking advantage of our money management expertise and variety of investment choices. That’s a good reason for consolidating your other eligible assets into an IRA with us.
3. The Directory of Intentional Communities lists more than 200 groups across the country organized around a variety of purposes, including environmentally aware living.
4. Sweeping changes occurred in demographics, economics, culture, and society during the last quarter of the 20th century. The nation aged, and more of its people gravitated to the Sunbelt. Sprawling “urban corridors” and “edge cities” challenged older central cities as sites for commercial, as well as residential, development. Rapid technological change fueled the growth of globalized industries, restructuring the labor force to fit a “postindustrial” economy.
5. “The most striking thing about Beijing, indeed about all of China, is that there are people everywhere. You need to imagine yourself in a never-ending Macy’s sale. There are lines to everything. You have to get in a line to find out which line to be in” (From Cheri Smith, *Suttertown News*, Sacramento, CA, March 19, 1987).
6. “Electrons, and protons, and some other particles (“charged” particles) exert a long-range force on one another. Like gravitation, this force is inversely proportional to the square of the distance between the particles; but unlike gravitation, this force is attractive or repulsive, depending on what particles are involved. Furthermore, this force is much stronger than the force of gravitation-between an electron and a proton, it is about 2×10^{39} times stronger. This force is called the electric force, and particles that exert these forces are said to carry electric charge.” (Ohanian, *Classical Electrodynamics*)
7. “I wish that everybody could feel the confidence of being alive in a fairly benign world, because I know that most of the advice and conclusions that scientists – scientists not being very scientific about it – have made concerning our planet and the perilous journey we’re on, is all bullshit. Because the place is very good at taking care of itself. We don’t have to take care of it. And only by being able to read and understand the articles in places like Science and Nature and Scientific American can you really come to that conclusion.

You're surrounded by this total barrage of faulty information that's been driven for economic reasons, information that's been basically made up for purposes of getting grants." (Kary Mullis, 1993 Nobel Laureate in chemistry, *Annals of Improbable Research*, vol. 5, no. 2, 1999)

8. Many of the world's greatest philosophers were bachelors. For instance, Descartes, Locke, Hume, and Kant were all unmarried.
9. The car's engine is running. When the driver pushes the start button, the starter motor's gear engages with the gear on the flywheel and cranks the engine, which causes it to start.
10. If Elon would read a book, he'd know we can't live on Mars.
11. Regular TV viewers overestimate both the rate of crime in their neighborhood and the personal danger to themselves and their families.
12. Although violent crime is actually decreasing, regular TV viewers overestimate their neighborhood crime rate by 150 percent and therefore misjudge personal danger to themselves and their families because local TV evening news regularly opens with graphic reports of mayhem and murder in familiar locations, making many believe that crime happens nightly outside their front door.